

CHAPTER – 1

ACCIDENTS IN INDIA

Introduction

This is the third edition of the report after the comprehensive revision of the proformae for data collection of 'Accidental Deaths & Suicides in India' completed in 2014 in consultation with States/UTs Police and others stakeholders.

An accident is an unplanned event that results in injury, deaths, damage to property or some other losses. For statistical purpose, Accidental Deaths have been classified as deaths that could be due to forces/factors of

the nature which have been termed as 'Accidental Deaths due to Forces of Nature' or could be due to deliberate or negligent conduct of human beings, which is termed as 'Other causes of Accidental Deaths'.

The rate of accidental deaths (per lakh of population) remained the same in 2016 as in 2015. The total number of accidental deaths in 2016 was 4,18,221 [Table-1(A)]. The incidence and rate of accidental deaths during 2006-2016 is presented in Table-1.1.

Table – 1 (A)
Percentage Change in Population, Number and Rate of Accidental Deaths over the Corresponding Previous Year during 2012 to 2016

Year	Mid-Year Projected Population* (In Lakh+)	Percentage Change in Population over the Previous Year	Percentage Change in Accidental Deaths over the Previous Year	Total Number of Accidental Deaths	Rate of Accidental Deaths	Percentage Change in Rate of Accidental Deaths over the Previous Year
(1)	(2)	(3)	(4)	(5)	(6)	(7)
2012	12133.7	0.3	1.0	394982	32.6	0.9
2013	12287.9	1.3	1.4	400517	32.6	0.0
2014	12440.4	1.2	12.8	451757	36.3	11.3
2015	12591.1	1.2	-8.5	413457	32.8	-9.6
2016	12739.9	1.2	1.2	418221	32.8	0.0

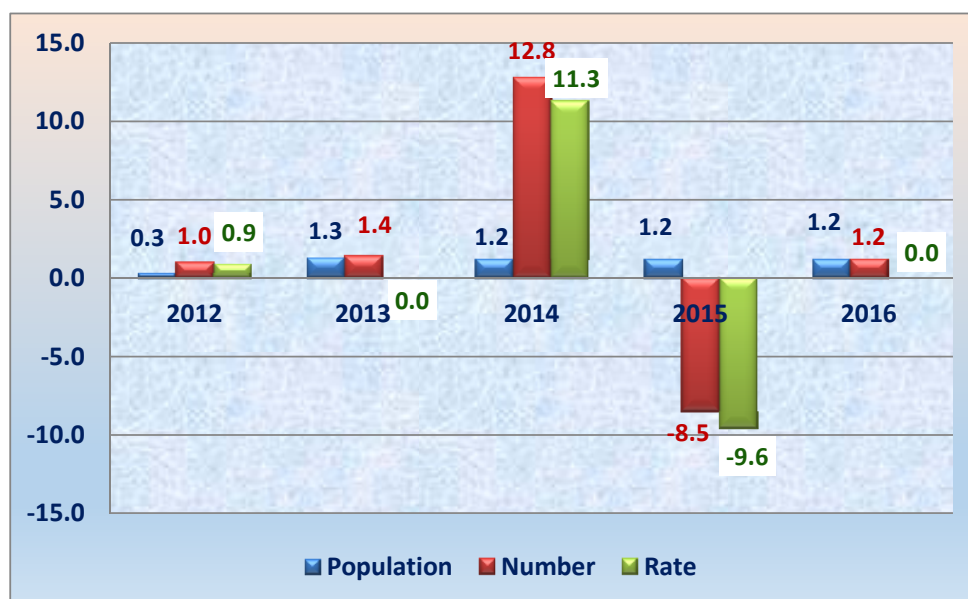
* — Source: The Registrar General of India (Provisional)

+ — One Lakh = 0.1 Million = 100,000

• As per data provided by States/UTs.

Figure 1.1

Percentage Change in Population, Number and Rate of Accidental Deaths during 2012 – 2016 (Over Corresponding Previous Year)



- As per data provided by States/UTs.

Out of 4,18,221 accidental deaths, 8,684 (2.1%) deaths were due to forces of nature and 4,09,537 (97.9%) deaths were due to other causes.

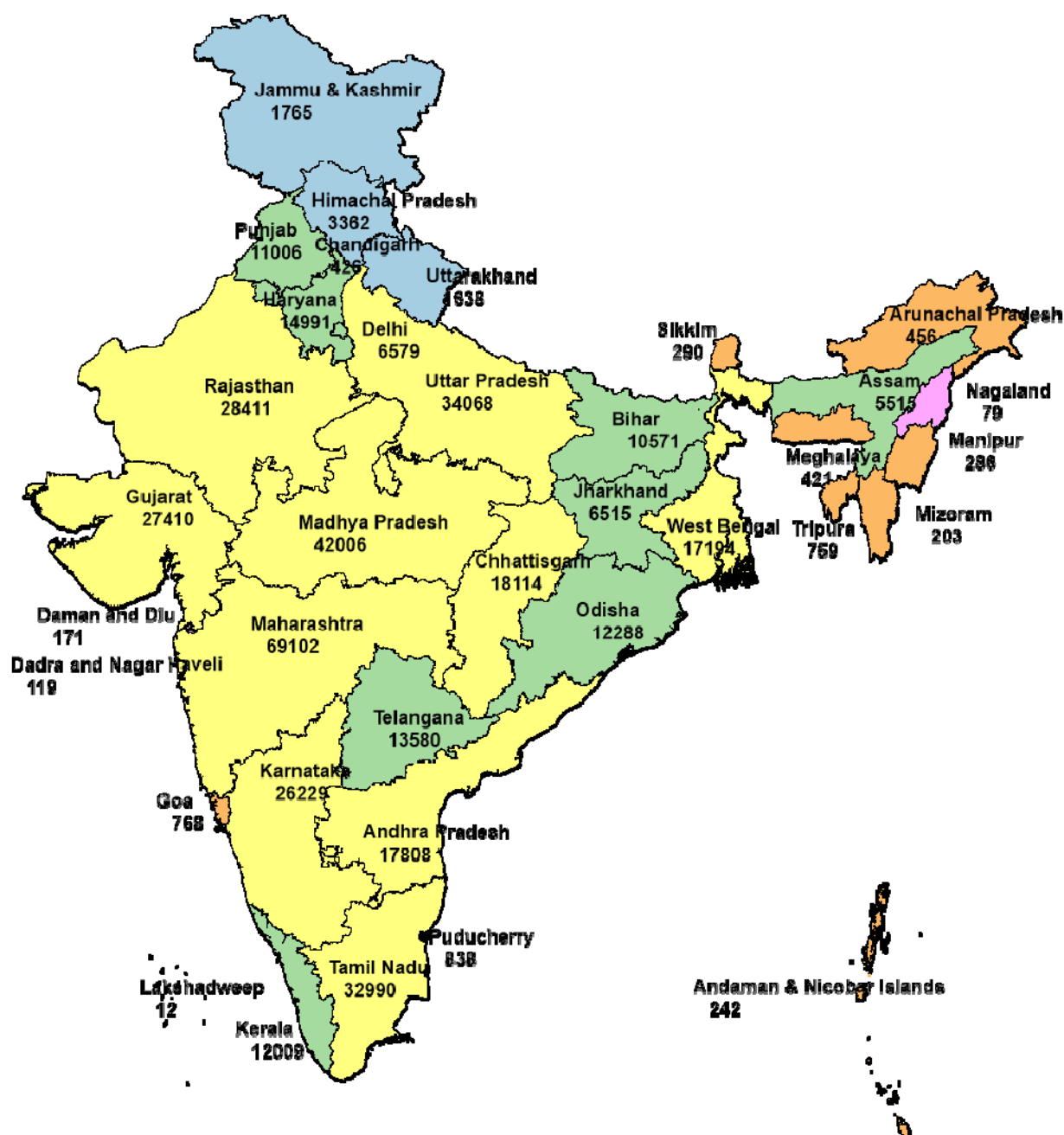
There was a decline of 17.4% (from 10,510 deaths in 2015 to 8,684 deaths in 2016) in deaths due to forces of nature and an increase of 1.6% (from 4,02,947 deaths in 2015 to 4,09,537 deaths in 2016) in deaths due to other causes during 2016 as compared to 2015 [Table-1(B)].

Table — 1 (B)
Number of Accidental Deaths due to Forces of Nature & Other Causes (2012 to 2016)

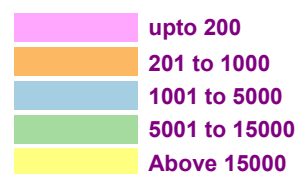
Year	Forces of Nature		Other Causes		Total Accidental Deaths	
	Number of Deaths	% change over previous year	Number of Deaths	% change over previous year	Number of Deaths	% change over previous year
(1)	(2)	(3)	(4)	(5)	(6)	(7)
2012	22,960	-3.1	3,72,022	1.3	3,94,982	1.0
2013	22,759	-0.9	3,77,758	1.5	4,00,517	1.4
2014	20,201	-11.2	4,31,556	14.2	4,51,757	12.8
2015	10,510	-48.0	4,02,947	-6.6	4,13,457	-8.5
2016	8,684	-17.4	4,09,537	1.6	4,18,221	1.2

- As per data provided by States/UTs.

STATE/UT - WISE ACCIDENTAL DEATHS DURING 2016



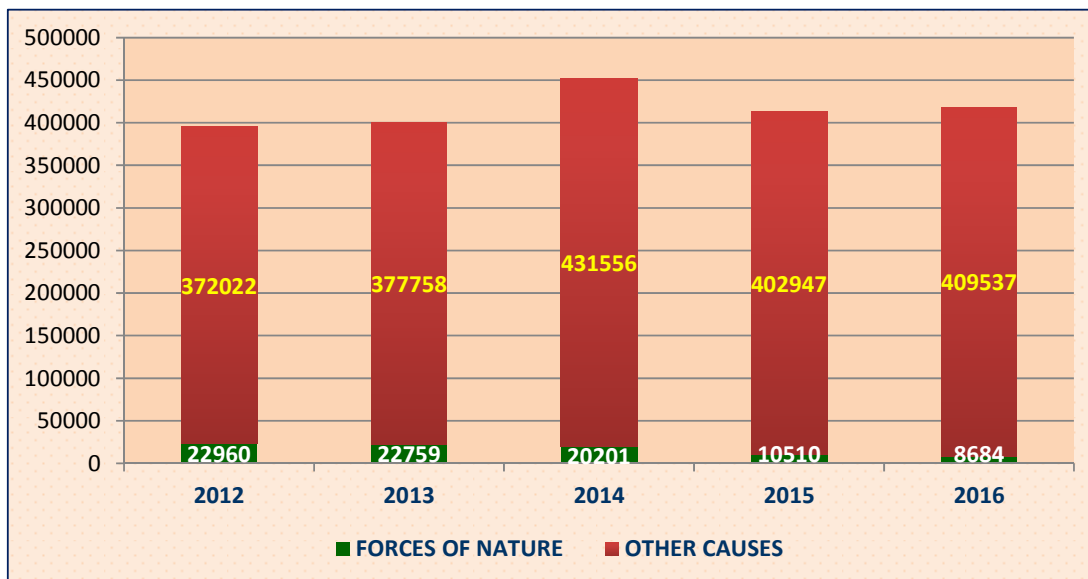
Accidental Deaths



• As per data provided by States/UTs.

Map Powered by DevInfo, UNICEF

Figure 1.2
Number of Accidental Deaths due to Forces of Nature and Other Causes during 2012-2016



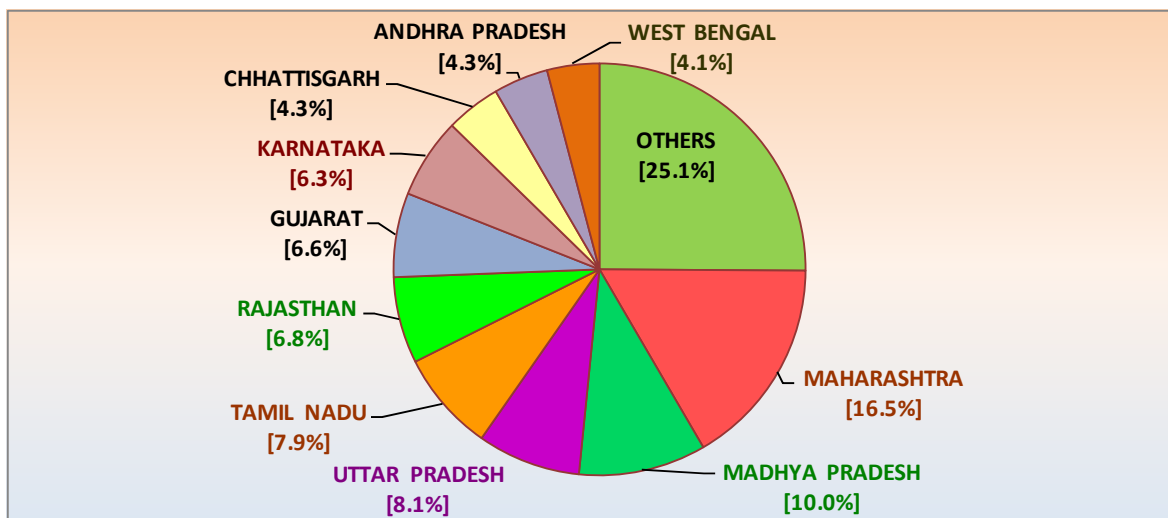
- As per data provided by States/UTs.

Number & Rate of Accidental Deaths in States, UTs and Cities

Out of 4,18,221 accidental deaths, like previous year, in 2016 also, Maharashtra with a population share of 9.5% has reported highest number of accidental deaths (69,102), contributing nearly one-sixth (16.5%) of total accidental deaths reported in the country. Uttar Pradesh, the most populous State in the

country with population share of 17.2% has accounted for 8.1% of accidental deaths in the country. The other States having higher percentage share in total accidental deaths were Madhya Pradesh (10.0%), Tamil Nadu (7.9%), Rajasthan (6.8%), Gujarat (6.6%), Karnataka (6.3%), Chhattisgarh (4.3%), Andhra Pradesh (4.3%) and West Bengal (4.1%) during 2016 [Table-1.2]. These States had nearly similar percentage share in 2015 also.

Figure 1.3
Percentage Share of Major States in Accidental Deaths during 2016



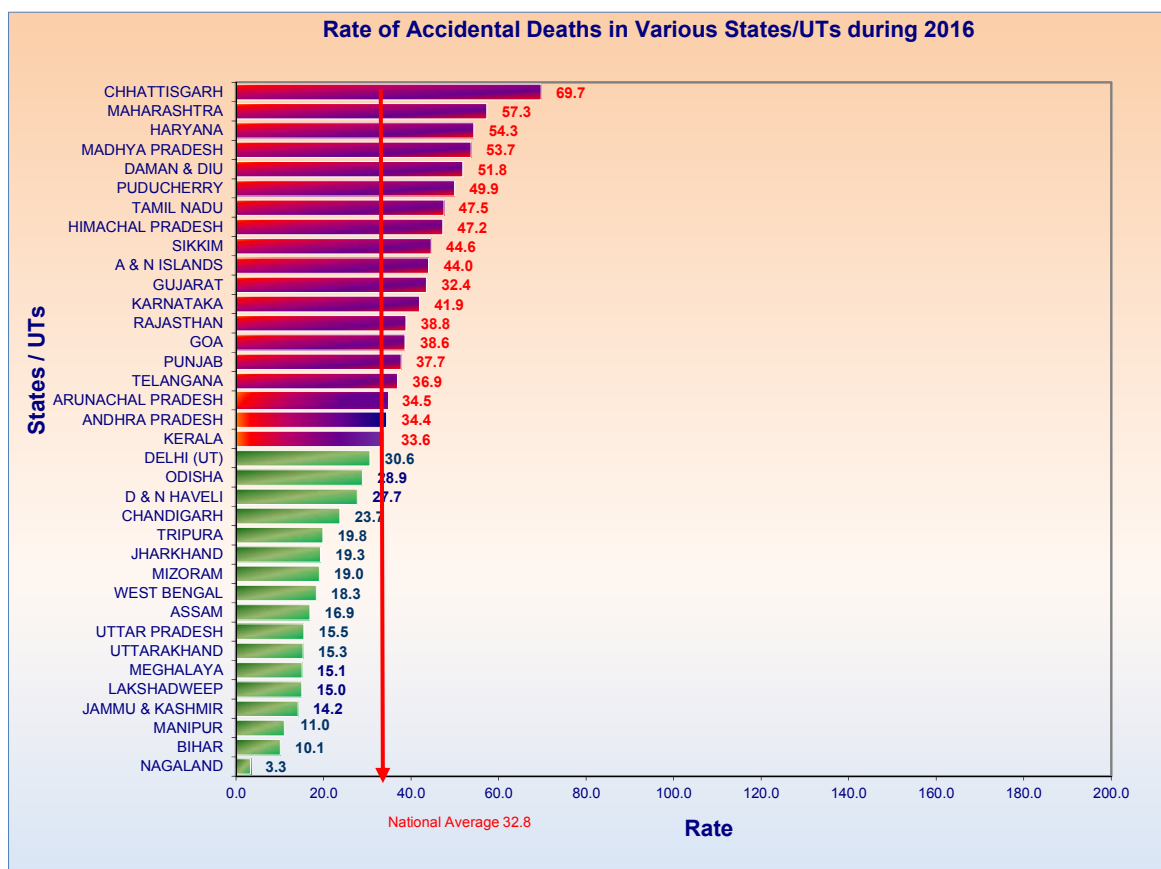
- As per data provided by States/UTs.

Other States/UTs include Arunachal Pradesh, Bihar, Assam, Chandigarh, Goa, Haryana, Jammu & Kashmir, Jharkhand, Kerala, Manipur, Mizoram, Meghalaya, Nagaland, Odisha, Punjab, Sikkim, Telangana, Tripura, Uttarakhand, A & N Island, D&N Haveli, Daman & Diu, Delhi, Lakshadweep and Puducherry

The highest **rate of accidental deaths** was reported from Chhattisgarh (69.7) followed by Maharashtra (57.3), Haryana (54.3), Madhya Pradesh (53.7) and Daman & Diu (51.8) against the national average rate of 32.8. 19 out of 36 States/UTs have reported

higher rates of 'Accidental Deaths' as compared to All India Average of 32.8 deaths per one lakh of population [Figure– 1.4].

Figure 1.4



- As per data provided by States/UTs.

The number of accidental deaths has increased by 1.2% at national level during 2016 as compared to 2015. The number of 'Accidental Deaths' due to forces of nature have decreased by 17.4% and deaths due to other causes have increased by 1.6% during 2016 over the year 2015 [Table-1.3].

Lakshadweep has reported the highest percentage increase (71.4%) in total accidental deaths in the year 2016 as compared to 2015 (from 7 in 2015 to 12 in 2016), followed by Chandigarh (60.2%), Sikkim (38.1%), Tripura (38.0%), A & N Islands (29.4%) and Assam

(27.3%). Uttarakhand has reported a decline of 40.4% in total accidental deaths in the year 2016 as compared to 2015 followed by Puducherry (30.6%), Nagaland (10.2%) and Uttar Pradesh (7.9%) [Table-1.3].

Gender-wise analysis reveals that female and male constitute 20.2% and 79.8% of total victims respectively. Age group of most of the victims was between 30 to 45 years. This group of people has accounted for 31.6% of all persons killed in accidents in the country during the year 2016. A total of 40,114 senior citizens 60 years & above of age) have also got killed in various accidents during 2016 [Table-1.4].

STATE/UT - WISE RATE OF ACCIDENTAL DEATHS DURING 2016



Note:

Rate of Accidental Deaths means Number of Accidental Deaths per 1,00,000 population.

Accidental Death Rate All India Average 32.8

	upto 10
	11 to 20
	21 to 30
	31 to 50
	Above 50

Map Powered by DevInfo, UNICEF

- As per data provided by States/UTs.

Causes of Accidental Deaths

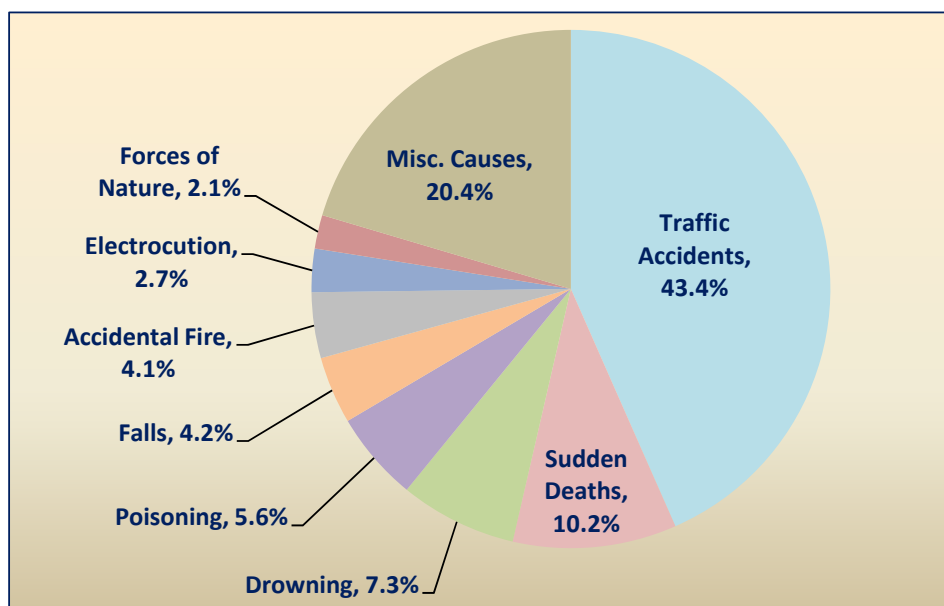
Accidental deaths have been grouped into two broad categories i.e. 'Deaths due to Forces of Nature' and 'Deaths due to Other Causes'. Classification of types and causes of accidental deaths has been done for statistical purpose only.

Under 'Causes Attributable to Forces of Nature', the deaths due to 'Lightning', and 'Forest Fire' have increased during the year 2016 over the previous year 2015 whereas the share of causes of deaths which *inter-alia* include 'Avalanche', 'Exposure to Cold', 'Tornado', 'Earth quake', 'Epidemic', 'Flood', 'Heat/Sun Stroke', 'Landslide' and 'Torrential Rain' have decreased in 2016 over 2015.

The deaths due to causes not attributable to forces of nature such as 'Collapse of Structure', 'Electrocution', 'Accidental Explosion', 'Falls', 'Mines or Quarry disaster', 'Traffic Accidents', 'Sudden Deaths', 'Deaths of Women during Pregnancy', 'Killed by Animals', and 'Drug Overdose' has increased whereas the deaths due to 'Air Crash', 'Ship Accidents', 'Drowning', 'Factory/ Machine Accidents', 'Accidental Fire', 'Firearm', 'Stampede', 'Deaths due to Consumption of Illicit/Spurious Liquor', 'Poisoning' and 'Suffocation' has decreased in 2016 as compared to the previous year 2015.

Cause-wise details of accidental deaths, rate and its percentage variation during 2016 over 2015 are presented in **Table-1(C)**.

Figure 1.5
Percentage Share of Various Major Causes of Accidental Deaths during 2016
(Forces of Nature and Other Causes)



- As per data provided by States/UTs.

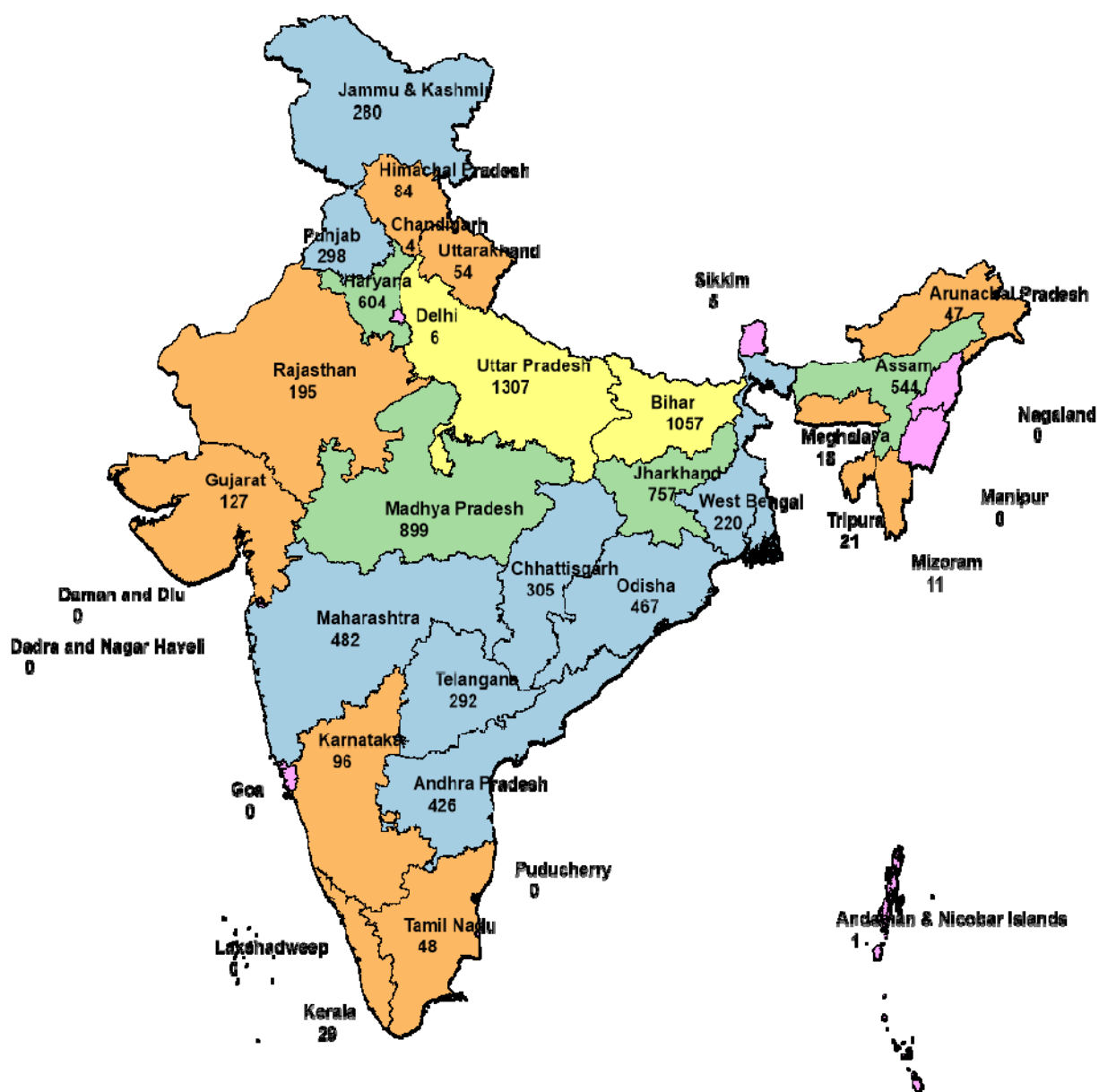
Misc. Causes include Collapse of Structure, Suffocation, Deaths of Women during Pregnancy, Killed by Animals, Deaths due to Consumption of Illicit/Poisonous Liquor, Accidental Explosion, Drug Overdose, Factory/Machine Accidents, Firearm, Mines or Quarry Disaster, Stampede, Air-Crash, Other than above & Unknown Causes.

Table-1(C)
Number and Share of Accidental Deaths due to Forces of Nature and Other Causes - 2015 & 2016

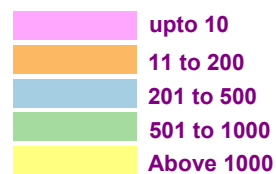
Sl. No.	Cause	2015		2016		% Variation during 2016 over 2015
		No.	% Share (w.r.t. All India)	No.	% Share (w.r.t. All India)	
(1)	(2)	(3)	(4)	(5)	(6)	(7)
A. FORCES OF NATURE						
1	Avalanche	38	0.4	19	0.2	-50.0
2	Exposure to Cold	1149	10.9	687	7.9	-40.2
3	Cyclone	15	0.1	15	0.2	-
4	Tornado	13	0.1	9	0.1	-30.8
5	Tsunami	0	0.0	0	0.0	-
6	Earthquake	92	0.9	3	0.0	-96.7
7	Epidemic	218	2.1	36	0.4	-83.5
8	Flood	846	8.0	775	8.9	-8.4
9	Heat/Sun Stroke	1908	18.2	1338	15.4	-29.9
10	Landslide	232	2.2	210	2.4	-9.5
11	Lightning	2641	25.1	3315	38.2	25.5
12	Torrential Rain	195	1.9	143	1.6	-26.7
13	Forest Fire	19	0.2	21	0.2	10.5
14	Causes other than above	3144	29.9	2113	24.3	-32.8
15	Total (A)	10510	100.0	8684	100.0	-17.4
B. OTHER CAUSES						
1	Air-Crash	23	0.0	2	0.0	-91.3
2	Ship Accidents	1	0.0	0	0.0	-100.0
3	Collapse of Structure	1885	0.5	1984	0.5	5.3
4	Drowning	29822	7.4	29721	7.3	-0.3
5	Electrocution	9986	2.5	11126	2.7	11.4
6	Accidental Explosion	831	0.2	852	0.2	2.5
7	Falls	16759	4.2	17278	4.2	3.1
8	Factory/Machine Accidents	695	0.2	685	0.2	-1.4
9	Accidental Fire	17700	4.4	16900	4.1	-4.5
10	Firearm	736	0.2	614	0.1	-16.6
11	Mines or Quarry Disaster	118	0.0	130	0.0	10.2
12	Traffic Accidents	177423	44.0	177904	43.4	0.3
13	Stampede	480	0.1	45	0.0	-90.6
14	Sudden Deaths	35023	8.7	41628	10.2	18.9
15	Deaths of Women during Pregnancy	1040	0.3	1131	0.3	8.8
16	Deaths due to Consumption of Illicit/Poisonous Liquor	1522	0.4	1054	0.3	-30.7
17	Killed by Animals	951	0.2	1063	0.3	11.8
18	Poisoning	26173	6.5	23094	5.6	-11.8
19	Suffocation	1437	0.4	1399	0.3	-2.6
20	Drug Overdose	750	0.2	778	0.2	3.7
21	Causes Not Known	15165	3.8	20979	5.1	38.3
22	Causes other than above	64427	16.0	61170	14.9	-5.1
	Total (B)	402947	100.0	409537	100.0	1.6
	Grand Total (A+B)	413457	100.0	418221	100.0	1.2

• As per data provided by States/UTs.

STATE/UT - WISE DEATHS DUE TO FORCES OF NATURE DURING 2016



Accidental Deaths



Map Powered by DevInfo, UNICEF

- As per data provided by States/UTs.

Accidental Deaths due to Forces of Nature

Cause - wise accidental deaths due to forces of nature are presented in **Table-1.5**. Out of 8,684 accidental deaths due to forces of nature, 38.2% deaths were due to 'Lightning' (3,315 deaths), 15.4% deaths due to 'Heat/Sun Stroke' (1,338 deaths), 8.9% deaths due to 'Flood' (775 deaths) and 7.9% deaths due to 'Exposure to Cold' (687 deaths) during the year 2016. 'Landslide' and 'Torrential Rain' caused 2.4% (210 deaths) and 1.6% (143 deaths) of total deaths due to forces of nature respectively during 2016.

The age group-wise distribution of accidental deaths by due to forces of nature is given in **Table-1.5**. As per age-wise classification of deaths due to forces of nature, 6.7% of such victims were below 14 years of age (586 out of 8,684 deaths), 7.3% in the age-group of 14 years & above to below 18 years (631 deaths), 20.3% in the age-group of 18 years to below 30 years (1,761 deaths), 28.2% in the age-group of 30 years to below 45 years (2,448 deaths) and 24.5% victims were in age group 45 years to below 60 years (2,128 deaths) during 2016. Senior citizens (60 yrs. and above) constituted 13.0% (1,130 deaths) of the total such victims.

State/UT - wise and gender - wise distribution of accidental deaths in the country by forces of nature along with their share in total deaths is presented **Table-1.8**. 'Lightning' caused 80.5%, 80.2% and 79.2%, of total such deaths in Odisha (376 out of 467 deaths), Karnataka (77 out of 96 deaths) and Tamil Nadu (38 out of 48 deaths) respectively. 'Heat/Sun Stroke' caused 74.0% (216 out of 292 deaths) of total accidental deaths due to forces of nature in Telangana. 100% (1 out of 1 death) of death in A & N Islands were due to 'Flood'. Similarly, 100% (4 out of 4 deaths) of deaths in Chandigarh were due to 'Exposure to Cold'. 'Landslide' caused 81.8%, 70.2%, 60.0% and 35.2% of total deaths in Mizoram (9 out of 11), Arunachal Pradesh (33 out of 47 deaths), Sikkim (3 out of 5 deaths) and Uttarakhand (19 out of 54 deaths) respectively during 2016.

Majority of accidental deaths due to 'Epidemic' and 'Exposure to Cold', were reported in Uttar Pradesh which accounted for 44.4% (16 out of 36 deaths), 27.4% (188 out of

687 deaths) of total such deaths reported in the country respectively during 2016. Majority of deaths due to 'Earthquake' were reported in Bihar which accounted for 66.7% (2 out of 3 deaths) of total such deaths during 2016.

Other Causes of Accidental Deaths

It is generally accepted that accidents due to 'Other Causes' are preventive and can be reduced by effective safety measures and safety consciousness.

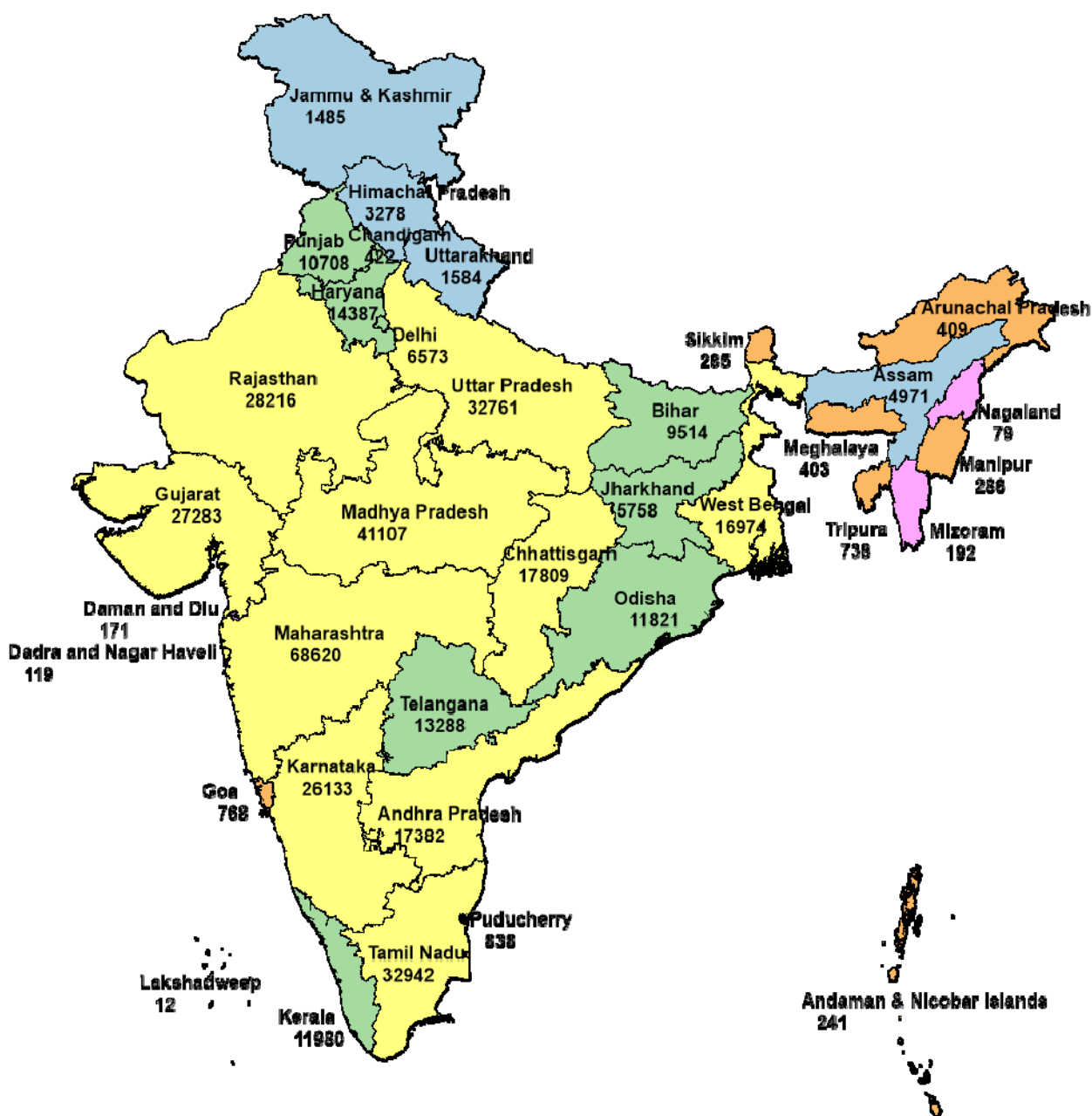
A total of 7,35,479 cases of 'Other Accidents' were reported in the country which resulted in 4,09,537 deaths and left 4,94,844 persons injured during 2016 [**Table-1.6**].

The cause-wise analysis of deaths revealed that majority of deaths were due to traffic accidents which accounted for 43.4% (1,77,904 out of 4,09,537 deaths) of total deaths due to 'Other Causes' during the year 2016. Besides, 'Sudden Deaths', 'Drowning', 'Poisoning' and 'Falls', caused 10.2% (41,628 deaths), 7.3% (29,721 deaths), 5.6% (23,094 deaths) and 4.2% (17,278 deaths) of total deaths due to other causes respectively [**Table-1(C)**].

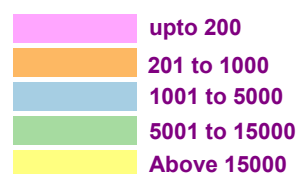
The age-group-wise distribution of accidental deaths in the country by various 'Other Causes' along with their share in total deaths is presented in **Table-1.7**. Majority of victims were in age groups 30 above - below 45 years and 18 above - below 30 years which accounted for 31.7% (1,29,701 deaths) and 26.9% (1,10,109 deaths) of total such deaths respectively.

State/UT-wise and gender-wise distribution of accidental deaths in the country by various other causes is presented in **Table-1.9**. Maharashtra with 9.5% (one-tenth) of country's population has reported 16.7% (68,620 out of 4,09,537 deaths) of accidental deaths due to other causes, Madhya Pradesh, Tamil Nadu, Uttar Pradesh, Rajasthan, Gujarat and Karnataka accounted for 10.0%, 8.0%, 7.9%, 6.9%, 6.7% and 6.4% of total such deaths respectively during the year 2016.

STATE/UT - WISE ACCIDENTAL DEATHS DUE TO OTHER CAUSES DURING 2016



Accidental Deaths



• As per data provided by States/UTs.

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State/UT-wise and Cause-wise details on 'Other Cause of Accidents' is presented in **Table 1.9**. State/UT-wise analysis reveals that a total of 23,093 out of 1,77,904 deaths due to traffic accidents were reported in Uttar Pradesh accounting for 13.0% of total such deaths in the country during 2016. A total of 12,370 out of 41,628 deaths were reported from Maharashtra under 'Sudden Deaths' accounted for 29.7% of total such deaths. Large number of deaths due to 'Drowning' were reported from Maharashtra (4,765 out of 29,721) accounting for 16.0% of such deaths. Maximum number of deaths due to 'Poisoning' were reported from Madhya Pradesh (5,225 out of 23,094) accounting for 22.6% of such deaths.

A total of 1,073 incidents of consumption of illicit/spurious liquor caused 1,054 deaths in the country during 2016. States where such deaths were reported include Madhya Pradesh (184 deaths) followed by Haryana (169 deaths), Chhattisgarh (142 deaths), Punjab (72 deaths) and Uttar Pradesh & Karnataka (68 deaths each) during 2016.

Fire Accidents

Considering the number of fire accidents in the country and potential of the damage that could be caused by such accidents, the Bureau has also collected data on fire accidents according to places of occurrence for appropriate preventive strategies viz. 'Fire in School Buildings', 'Fire in Commercial Buildings', 'Fire in Residential / Dwelling

Buildings', 'Fire in Government Buildings', 'Fire in Mines', 'Fire in Factories', 'Accidental Fire in Trains' and 'Accidental Fire in Vehicles' [**Table-1.10**].

A total of 16,695 cases of fire accidents were reported in the country during 2016, showing a decrease of 9.5% during 2016 over 2015 (18,450 cases). 16,695 incidents of fire accidents caused injuries to 998 persons and 16,900 deaths during 2016. The cause-wise analysis of fire accidents revealed that 50.2% of total deaths (8,478 out of 16,900) due to fire accidents were reported in residential/dwelling buildings during 2016 [**Table 1.11**].

Large number of cases of fire accidents were reported in Maharashtra (3,063 out of 16,695), accounting for 18.3% of total such cases during 2016 [**Table-1.11**].

The State/UT-wise analysis of place of fire accidents revealed that 24 States & UTs have reported 50.0% or more deaths due to 'Fire in residential or dwelling building' during 2016. These States/UTs were Mizoram, Nagaland, Sikkim, Tripura, Chandigarh UT (100% each) Arunachal Pradesh (90.9%), Uttarakhand (89.3%), Goa & Telangana (82.6% each), Bihar (80.5%), and Tamil Nadu (80.4%) [**Table-1.11**].

Accidents in Cities

The term 'Mega City' refers to cities which have a population of over 10 lakh or more. As per the Population Census 2011, the status of mega city has been accorded to 53 cities.

A total of 69,400 accidental deaths were reported in 53 mega cities during 2016. Maximum numbers of accidental deaths (8,713 deaths) were reported in Mumbai accounting for 12.6% of total deaths, followed by Pune (5,659), Delhi City (5,468), Chennai (4,081), Bengaluru (3,756) and Jaipur (3,212).

Average rate of accidental deaths in these 53 mega cities was higher (43.2) than all India rate of accidental deaths (32.8). The rate of accidental deaths was highest in Bhopal (159.5) followed by Asansol (116.9), Pune (112.1), Jaipur (104.6), Raipur (90.0) and Nasik (88.7) as compared to average rate for the cities (43.2) [Table-1.2].

Out of 69,400 accidental deaths in 53 mega cities; 495 and 68,905 deaths were due to forces of nature and other causes of accidents respectively during 2016. A decrease of 70.7% was observed in deaths due to forces of nature and an increase of 1.8% was witnessed in deaths due to 'Other' causes during 2016 over 2015 respectively.[Table-1.3]

Kolkata, the third largest city in terms of population (8.8% of the total population of all mega cities) among 53 cities, reported the lowest share of deaths in cities accounting for 1.1% of total accidental deaths in 53 mega cities during 2016. The comparative reporting of 'Accidental Deaths' during 2016 in the top four-metropolitan cities is presented in Table-1(D).

Accidental Deaths due to Forces of Nature in Cities

A total of 495 deaths due to forces of nature were reported in these 53 mega cities during 2016. Among specified causes of accidents due to forces of nature in 53 cities, maximum deaths were reported due to 'Exposure to cold' followed by 'Heat/Sun Stroke', accounting for 26.7% and 21.4% of total such accidental deaths during 2016. All deaths in Vijayawada (11 deaths) were due to 'Heat/Sun Stroke' during 2016. All 4 deaths in Chandigarh City, 75.9% (22 deaths out of 29 deaths) in Allahabad and 62.5% (5 deaths out of 8 deaths) in Agra were due to 'Exposure to Cold'.

Accidental Deaths due to Other Causes in Cities

A total of 1,37,177 cases of 'Other' accidents were reported in 53 mega cities which caused injuries to 74,674 persons and 68,905 deaths during 2016. The cause-wise analysis of deaths due to other causes revealed that 26.7%, 5.8%, and 3.9% deaths in 53 cities were due to 'Traffic Accidents', 'Falls' and 'Drowning' respectively.

Among 53 cities, Mumbai city accounted for 12.6% of total deaths reported under 'Other' accidents followed by Pune and Delhi City which accounted for 8.2% and 7.9% respectively during 2016.

"Data on City-wise information is available on our website under 'Additional Tables for ADSI'"

Table – 1 (D)
Rate of Accidental Deaths in Metropolitan Cities during 2016

Sl. No.	City	Population (In Lakhs)	Share of Population In Cities (Percent)	Share of Deaths In Cities (Percent)	Rate (No. of Deaths Per One Lakh Population)
(1)	(2)	(3)	(4)	(5)	(6)
1.	Chennai	87.0	5.4	5.9	46.9
2.	Delhi	163.2	10.1	7.9	33.5
3.	Kolkata	141.1	8.8	1.1	5.3
4.	Mumbai	184.1	11.4	12.6	47.3

- As per data provided by States/UTs.
